Running schema and data migrations is one of those administrative processes described by the 15-Factor methodology introduced in chapter 2. In this case, the strategy adopted for managing such a process was to embed it in the application itself. By default, it's activated during the application startup phase. Let's see how we can implement it for Catalog Service.

Open your Catalog Service project (catalog-service) and add a dependency on Flyway in the build.gradle file. Remember to refresh or reimport the Gradle dependencies after the addition.

Listing 5.20 Adding a dependency on Flyway in Catalog Service

```
dependencies {
  ...
  implementation 'org.flywaydb:flyway-core'
}
```

In the next section, you'll learn how to create your first migration to initialize the database schema.

5.4.2 Initializing a database schema with Flyway

The first database change you'll apply is usually to initialize the schema. So far we've been relying on the built-in data source initialization capabilities offered by Spring Boot and providing a schema.sql file with the SQL statements to run. Now we can initialize the schema using an SQL Flyway migration.

First, delete the schema.sql file and remove the `spring.sql.init.mode` property from the application.yml file in your Catalog Service project.